8 Collisions

Introduction

Energy and momentum are two physical quantities that define a mechanical system. The principle of energy conservation and momentum conservation (in case of no external forces) or change in these quantities allows us to analyze systems without the need to solve equations of motion (Newton's 2nd law). Elastic and inelastic collisions are performed with two carts of different masses moving on a levelled 1.2 m aluminum track. The carts are equipped with magnetic bumpers on one end and Velcro bumpers on their other end. The magnetic bumpers are used in elastic collisions and the Velcro bumpers are used in the completely inelastic collision. Plungers at the end of the cart can be used to mimic the event of explosion. Additional mass can be added on top of the carts to explore collisions of objects with unequal masses.

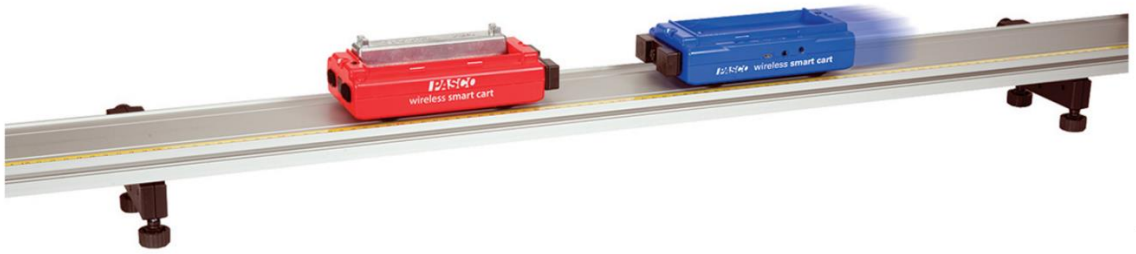


Figure 1. Collisions experimental setup

Theoretical model

The kinetic energy of an object of mass m moving with velocity $\vec{v}$ is

$$K = \frac{1}{2}mv^2 \qquad \text{(eq. 1)}$$

The total kinetic energy of a system of several objects is the **algebraic sum** of the individual kinetic energies of each object.

The moving object also has (linear) momentum $\vec{p}$ defined as

$$\vec{p} = m\vec{v} \qquad \text{(eq. 2)}$$

The total momentum of a system of several objects is the **vector sum** of the individual momenta of each object.

The definition of momentum allows us to write Newton's 2nd law for an object:

$$\vec{F}_{net} = \frac{d\vec{p}}{dt} \qquad \text{(eq. 3)}$$

That is, the net force (vector sum of all forces) acting on an object equals the time rate of change of momentum of the object.

For a system of several objects, the objects can interact with each other by collisions and exchange of momentum, these constitute internal forces that do not change the total momentum of the system.

The system can also interact with the environment if the environment applies external forces on the objects in the system. The total external force applied to the system results in change in the total momentum of the system.

$$\vec{F}_{ext,tot} = \frac{d\vec{p}_{tot}}{dt} \qquad \text{(eq. 4)}$$

That is, for a system (or an object) that is not influenced by external forces, $\vec{F}_{ext,tot} = 0$, the total momentum is not changing with time, or in other words – total momentum is conserved.

$$\frac{d\vec{p}_{tot}}{dt} = 0 \quad \rightarrow \quad \vec{p}_{tot,before} = \vec{p}_{tot,after} \qquad \text{(eq. 5)}$$

If the momentum of the system is conserved and the internal forces in the system are conservative, then the total kinetic energy of the system is also conserved. These type of collisions are called ***elastic collisions.*** However, if the internal forces in the system are non-conservative, then the total kinetic energy of the system is not conserved (even if there are no external forces) because mechanical energy can be lost as heat. We refer to this as ***inelastic collisions,***

A third type of interaction is an **explosion**. When an object explodes into several smaller objects internal stored energy is released. In this process the kinetic energy is not conserved and the final energy is greater than the initial energy.

Consider two objects of masses, m_1 and m_2 move towards each other with known velocities v_1 and v_2, respectively. The masses they collide elastically in 1 dimension.

First, we want to find a general expression for the magnitude and direction of velocities after the collision, v_1' and v_2'.

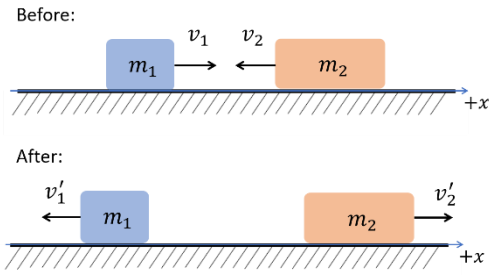


Figure 2. Elastic collision of two masses in 1D

Conservation of momentum gives

$$m_1 v_1 - m_2 v_2 = m_2 v_2' - m_1 v_1' \qquad \text{(eq. 6)}$$

Conservation of energy gives

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2 \qquad \text{(eq. 7)}$$

Rearrange both equations so that the velocities of m_1 on one side and the velocities of m_2 on the other side, then divide the equations and after some algebra we get the velocities of the masses after the collision:

$$v_1' = \frac{m_2 - m_1}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$
$$v_2' = \frac{2m_1}{m_1 + m_2} v_1 + \frac{m_1 - m_2}{m_1 + m_2} v_2 \qquad \text{(eq. 8)}$$

The complicated motion of a system of particles can be simplified if we define a special point of the system – the center of mass. The motion of a system's center of mass behaves as if all the mass were concentrated there and all the external forces act at that point.

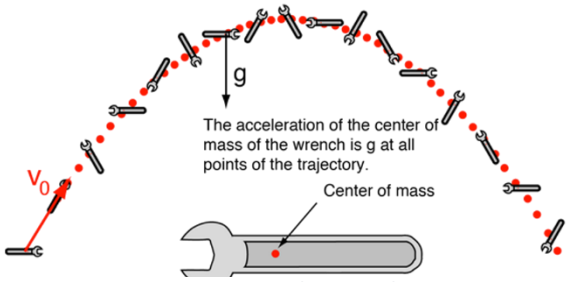


Figure 3. Motion of center of mass

Recalling Newton's 2nd law for a system of particles in terms of momentum (eq. 4), Since the total momentum is $\vec{p}_{tot} = \sum_i m_i \vec{v}_i$ and the total mass is $M_{tot} = \sum_i m_i$ we get Newton's 2nd law for the acceleration of the center of mass $\vec{a}_{CM}$ (show derivation):

$$\vec{F}_{ext,tot} = M_{tot}\vec{a}_{CM} \qquad \text{(eq. 9)}$$

where

$$\vec{a}_{CM} = \frac{\sum_i m_i \vec{a}_i}{\sum_i m_i} \qquad \text{(eq. 10)}$$

Next, we can integra Eq. 10 by time to get the velocity of the center of mass:

$$\vec{v}_{CM} = \frac{\sum_i m_i \vec{v}_i}{\sum_i m_i} \qquad \text{(eq. 11)}$$

Which is related to the total momentum of the system by

$$\vec{p}_{tot} = M_{tot}\vec{v}_{CM} \qquad \text{(eq. 12)}$$

Again, we can integrate by time Eq. 11 to find the position of the center of mass

$$\vec{r}_{CM} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} \qquad \text{(eq. 13)}$$

You can think of the location of the center of mass as the weighted average position of all the masses in the system.

Preparation Questions

1. Prove that for a system of N masses where only internal forces present (such as collisions) and there are no external forces (such as gravity) then the total momentum of the system is conserved.
2. Make the full derivation for the final velocities of elastic collision of two masses. That is, do all the missing steps from Eq. 16 and Eq. 17 to derive Eq. 18.
3. Check eq. 18 when m_2 is initially at rest in the following limiting cases: Equal masses, $m_1 = m_2$; A massive target, $m_1 \ll m_2$; A massive projectile, $m_1 \gg m_2$. Provide an intuitive explanation for each result.
4. Explore the motion of the center of mass in a collision of two carts of equal masses that travel towards each other with equal speed where the initial distance between the carts is L in the two following cases:
   a. Elastic collision with no external forces.
   b. Elastic collision with kinetic friction force with friction coefficient μ.

   Find the motion of the center of mass by (1) calculating the motion of each cart. (2) Use the kinematics of the center of mass motion.
5. Following part (4): Will your answer change if the collision is inelastic?

Experimental procedure and data analysis

Part A: Setting up the system:

1. Verify that the track is stable and levelled.
2. **Weigh the mass of each cart and the mass of the two extra masses and the 1 kg weight.**
3. In PASCO Capstone, click on 'file' and open an experiment named 'CO exp setup.cap' that you have downloaded from Moodle.
4. Turn on both cars and connect to the computer via Bluetooth in Hardware Setup in Capstone.
5. **To make the velocities of both carts positive to the right:**
   With the red cart to the left of the blue cart, face both carts with their magnetic bumpers to the right. Start recording and push both carts to the right. Both velocities should be positive. This establishes the coordinate system to have positive x to the right for both carts.
6. **In Capstone, on the tools ribbon (on the left side), open the calculator and create these calculations:**
   m1=enter the blue cart mass
   m2=enter the red cart mass
   v1=[Velocity, Blue m/s]
   v2=[Velocity, Red m/s]
   Ptot=m1*v1+m2*v2
   Ktot=0.5*m1*v1^2+0.5*m2*v2^2
7. Go to the page named "collisions". In each window left press the central icon and choose graph. Create 4 graphs: $v_1(t), v_2(t), P_{tot}(t)$ and $K_{tot}(t)$.

Part B: Elastic collisions – different cases:

1. **Elastic collisions – equal cart masses:**
   1.1 Place the red and blue carts (no cart mass bars) at rest on the track at some distance apart, with the magnetic bumpers facing each other, as shown in figure 4.

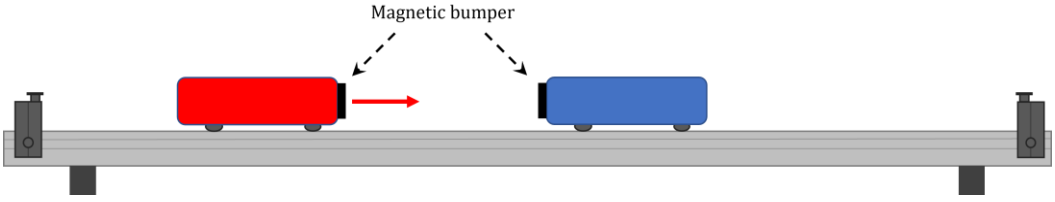


Figure 4. Elastic collision with two equal cart masses.

   1.2 Start recording and give the red cart a push toward the blue cart. Stop recording before either cart reaches the end of the track.

1.3 **Save the experiment data and the 4 graphs (velocities the carts, total momentum and total energy as function of time) as images.**

1.4 Find the velocity of the red and the blue carts just before and just after the collision. It may be helpful to expand the graph, to see just that area you are interested in.

Use data selector, data highlighter and statistics to find for each cart its velocity, momentum and kinetic energy before and after the collision.



1.5 Similarly, calculate the total momentum and total kinetic energy before and after the collision.

2. **Elastic collision – heavy target:**

2.1 Place the two mass bars and the 1 kg weight on the blue cart, as shown in Figure 5 below.

2.2 **Repeat the steps you followed in part B.1, but now with unequal cart masses. Don't forget to change the mass of the blue cart in the calculations!**

2.3 Save the experiment data in a separate file.

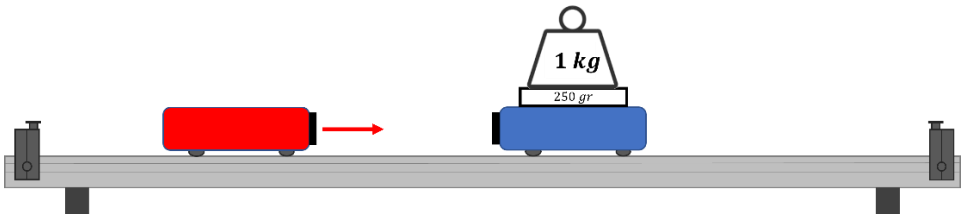


Figure 5. Elastic collision with unequal cart masses.

3. **Elastic collision – heavy projectile:**

3.1 Place the two mass bars and the 1 kg weight on the red cart, as shown in Figure 6 below.

3.2 **Repeat the steps you followed in part B.1, but now with unequal cart masses. Don't forget to change the mass of the red cart in the calculations!**

3.3 Save the experiment data in a separate file.

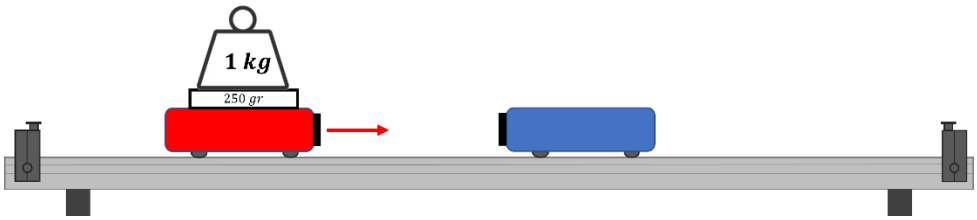


Figure 7. Elastic collision with unequal cart masses.

Part C: Completely Inelastic Collision

1. Place the red and blue carts (no cart mass bars) at rest on the track at some distant with the Velcro bumpers facing each other.

When the carts collide, they will hit Velcro bumper to Velcro bumper and stick together, resulting in an inelastic collision.

2. **Because we are reversing the carts, to keep the same frame of reference, open the "Data Summary" and click on the properties button next to the Red Smart Cart Position Sensor and select Change Sign. Do the same for the blue cart as well.**
3. Start recording and give the red cart a push toward the blue cart. Stop recording before either cart reaches the end of the track.
4. **Record the run number. Save the experiment data in a separate file and the 4 graphs as images.**
5. **Calculate for each cart its velocity, momentum and kinetic energy before and after the collision. Also, calculate the total momentum and total kinetic energy before and after the collision.**

<u>Part D: Explosion</u>

1. **Depress the plunger on one cart to position #2. Place the two carts in contact with each other (the Velcro bumpers in touch) in the center of the track.**
2. Start recording and tap the trigger release to launch the carts. Hitting the trigger with a mass bar works well. Stop recording before either cart reaches the end of the track.
3. **Record the run number. Save the experiment data in a separate file and the 4 graphs as images.**
4. **Calculate for each cart its velocity, momentum and kinetic energy before and after the collision. Also, calculate the total momentum and total kinetic energy before and after the collision.**

<u>Part E: Motion of center of mass</u>

1. **In Capstone, on the tools ribbon (on the left side), open the calculator and calculations to find the total force on both carts, the velocity and position of the center of mass:**

   Ftot=

   Vcm=

   Xcm=
2. Go to the page named "Forces". In each window left press the central icon and choose graph. Create 3 graphs: the force on the red cart, the force on the blue cart and the total force.
3. **<u>Use the data selector to choose the data of one of the collision cases you have tested in the previous parts.</u> Save the 3 graphs as images. What are your conclusions from the graphs?**



4. Go to the page named "CM". In each window left press the central icon and choose graph. Create 4 graphs: velocity of red cart, velocity of blue cart, the velocity and position of the center of mass.

5. **Use the same data for the same collision case you used for the force graphs. Save the graphs as images. What are your conclusions from the graphs?**
6. In general, what did you learn about conservation of momentum and kinetic energy in different types of collisions?
    a) Was momentum conserved for all types of collisions?
    b) Was total velocity conserved for all types of collisions?
    c) Was energy conserved for all types of collisions? Where did the extra kinetic energy come from in the explosions? What happens to the initial kinetic energy that is lost in a collision?

References

[1] Halliday, D., Resnick, R., & Walker, J. (2013). Fundamentals of physics 9th edition. John Wiley & Sons, pp. 73-75 and the example in 201-228.

[2] Young, H. D., Freedman, R. A., Sandin, T. R., & Ford, A. L. (2012). University physics 13th edition. Reading, MA: Addison-Wesley. pp. 241-266.